CHAPTER 6: Filesystem

Question 1: Environment Variables in the Terminal

Environment variables are dynamic, named key—value pairs maintained by the operating system shell that define system behavior and environment configuration for running processes. Their primary ~~propose~~ purpose is to provide universal configuration settings—such as default paths, user identity, locale, and terminal type—without hardcoding them into individual programs.

Question 2: The PATH variable, search mechanics, empty PATH, and /bin/ls

What is PATH? An environment variable containing a colon—delimited list of directory paths (e.g., /usr/local/bin:/usr/bin:/bin).

How the shell uses it: When you type a bare command name (like ls), the shell searches through each listed directory from left to right. As soon as it finds an executable matching that name, it launches it.

What happens if PATH is empty? The shell no longer knows where standard binaries live. Typing ls, grep, or cat returns command not found.

Why /bin/ls still works: When you provide an absolute path beginning with /, the shell bypasses PATH lookup entirely and invokes the binary file directly at that exact filesystem location.

What which is used for: It locates and displays the exact filesystem path of the executable that would run if typed into the terminal (e.g., which python3 → /usr/bin/python3).